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Review

Thermodynamic reassessment of the Sm–Ni binary system

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ABSTRACT

Based on the available experimental data of phase equilibria and thermodynamic properties from the literature, the Sm–Ni binary system has been thermodynamically assessed using the CALPHAD (CALculation of PHAse Diagrams) method. The solution phases, Liquid, FCC_A1, RHOMB, HCP_A3 and BCC_A2, were modeled as substitutional solution phases, of which the excess Gibbs energies were formulated with Redlich–Kister polynomials. All intermetallic phases Sm_3Ni, Sm_7Ni_3, Sm_3Ni_2, SmNi, $SmNi_2$, $SmNi_3$, Sm_2Ni_7, Sm_5Ni_{19}, $SmNi_5$, and Sm_2Ni_{17} were described as stoichiometric compounds. Subsequently, a set of self-consistent thermodynamic parameters describing various phases in this binary system has been obtained. The calculated results reproduce well the corresponding experimental data.

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1. Introduction

The intermetallic compounds formed by rare earth (RE) elements and transition metals (TM) are of particular interest regarding to their potential usage as high value functional materials, such as permanent magnets [1,2] and hydrogen storage materials (reversible absorption of a large quantity of hydrogen gas at room temperature and nearly at atmospheric pressure) [3,4]. Moreover, many ternary Al–TM–RE systems form amorphous alloys with interesting mechanical properties [5,6] and some rare-earth/transition metal oxides are candidate materials for solid oxide fuel cells [7]. Furthermore, the rare-earth/3d transition metal intermetallic compounds were suggested as being promising candidates for room temperature magnetic refrigeration based on magneto-caloric effect (MCE) [8–10] who has attracted more attention for improved energy efficiency and environmental friendliness compared with traditional compression/expansion gas

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refrigeration [11,12].

On the other hand, the knowledge of thermodynamic data and phase diagrams is essential for developing and checking models of phase transformations, coupling thermodynamics with kinetics [13]. The purpose of the present work is (1) to evaluate recent experimental phase diagram and his relative thermodynamic data and (2) to provide a set of self-consistent parameters for calculation of the phase equilibria and thermodynamic properties in the Sm–Ni binary system using the CALPHAD (CALculation of PHAse Diagram) method [14] and the Thermo-calc software package [15].

2. Thermodynamic models

2.1. Pure elements

The stable forms of the pure elements at 298.15 K and 1 bar were chosen as the reference states of the system. For the thermodynamic functions of the pure elements in their stable and metastable states, the phase stability equations compiled by Dinsdale [16] and adopted by the SGTE (Scientific Group Thermodata Europe) were used in the present work:

$$^{0}G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where H_i^{SER} (298.15 K)is the molar enthalpy of the so-called "Standard Element Reference" (SER), i.e., the enthalpies of the pure elements in their defined reference state at 298.15 K and 1 bar; T is the absolute temperature; $G_i^{\phi}(T)$ is the absolute molar Gibbs energy of the element i (i = Sm and Ni) with structure ϕ in a non magnetic states.

In thermodynamic study, absolute energy is not of importance, so the relative value of Gibbs energy $^{0}G_i^{\phi}(T)$ is adopted in CALPHAD approach. The Gibbs energy of the element i (i = Sm or Ni) $^{0}G_i^{\phi}(T)$, in its SER state is denoted by $GHSER_i$, i.e.

$$GHSER_{Sm} = {}^{0}G_{Sm}^{rhomb}(T) = G_{Sm}^{rhomb}(T) - H_{Sm}^{SER} \quad (298.15K) \quad (2)$$

$$GHSER_{Ni} = {}^{0}G_{Ni}^{fcc}(T) = G_{Ni}^{fcc}(T) - H_{Ni}^{SER} \quad (298.15K) \quad (3)$$

2.2. Solution phases

The substitutional solution model was employed to describe the solution phases including Liquid, FCC_A1, RHOMB, BCC_A2 and HCP_A3. The molar Gibbs energy of the solution phase ϕ (ϕ = Liquid, FCC_A1, RHOMB, HCP_A3 and BCC_A2) can be expressed as:

$$G_m^{\phi} = x_{Ni}{}^{0}G_{Ni}^{\phi} + x_{Sm}{}^{0}G_{Sm}^{\phi} + RT(x_{Ni}\ln x_{Ni} + x_{Sm}\ln x_{Sm}) + {}^{E}G_m^{\phi} + {}^{mg}G_m^{\phi} \quad (4)$$

where G_m^{ϕ} is the molar Gibbs energy of a solution phase ϕ; $^{0}G_i^{\phi}$ is the molar Gibbs energy of the element i (i = Sm, Ni) with the structure ϕ in a non-magnetic state; x_i the mole fraction of component i, R gas constant, T temperature; $^{E}G_m^{\phi}$ the excess Gibbs energy, and $^{mg}G_m^{\phi}$ is the magnetic contribution to the Gibbs energy which will be discussed in section 2.4. The excess Gibbs energy of phase ϕ can be expressed by the Redlich–Kister polynomials [17] as:

$$^{E}G_m^{\phi} = x_{Ni}x_{Sm}\sum_j {}^{j}L_{Ni,Sm}^{\phi}(x_{Ni} - x_{Sm})^j \quad (5)$$

here $^{j}L_{Ni,Sm}^{\phi}$(j = 0, 1, 2, …) is the interaction parameter between elements Sm and Ni and is formulated as temperature dependent:

$$^{j}L_{Ni,Sm}^{\phi} = a_j + b_jT + c_jTLnT + d_jT^2 + e_jT^3 + f_jT^{-1} + g_jT^7 + hjT^{-9} \quad (6)$$

Where a_j, b_j, c_j, d_j, e_j, f_j, g_j and h_j are model parameters to be optimized. In most cases, only the two first terms of the above equation are used.

2.3. Intermetallic compounds

All intermetallic compounds, Sm_3Ni, Sm_7Ni_3, Sm_3Ni_2, SmNi, $SmNi_2$, $SmNi_3$, Sm_2Ni_7, Sm_5Ni_{19}, $SmNi_5$, and Sm_2Ni_{17} were treated as stoichiometric phases in the Sm–Ni binary system because no available experimental data are reports homogeneity range for these compounds. The Gibbs energy of a Sm_ANi_B compound is given as:

$$^{0}G_m^{Sm_ANi_B} = \frac{A}{A+B}{}^{0}G_{Sm}^{rhomb} + \frac{B}{A+B}{}^{0}G_{Ni}^{fcc} + a + bT + {}^{mg}G_m^{Sm_ANi_B} \quad (7)$$

where $^{0}G_{Sm}^{rhomb}$ and $^{0}G_{Ni}^{hfcc}$ are the Gibbs energies of the respective pure elements Sm and Ni in the non-magnetic rhomb and fcc structure respectively. The parameters a and b were evaluated in the present work. $^{mg}G_m^{Sm_ANi_B}$ is the magnetic contribution to the Gibbs energy discussed in Section 2.4.

2.4. Magnetic contribution to the Gibbs energy

The magnetic contribution to the Gibbs energy $^{mg}G_m$ can be described as:

$$^{mg}G_m = RTln(\beta + 1)f(\tau) \quad (8)$$

where τ is defined as T/T_c with T_c being the critical temperature for magnetic ordering, i.e. the curie temperature (Tc) for ferromagnetic ordering and the Néel temperature (T_N) for antiferromagnetic ordering; β is a quantity related to the total magnetic entropy and is equal to the Bohr magnetic moment per mole of atoms in most cases; $f(\tau)$ represents the polynomials given by Hillert and Jarl [18] based on the magnetic specific heat of iron, i.e.for $\tau < 1$

$$f(\tau) = 1 - \frac{79\tau^{-1}/140p + 474/497(1/p - 1)(\tau^3/6 + \tau^9/135 + \tau^{15}/600)}{D} \quad (9)$$

Table 1
Curie temperature and Bohr magnetic for Sm–Ni intermetallic compounds.

Intermetallic compound	SmNi	$SmNi_2$	$SmNi_3$	$SmNi_5$	Sm_2Ni_{17}
T_C (K)	45 [19]	22.5 [20]	85 [21]	27.5 [22]	186 [23]
β ($\mu_{B/formula\ unit}$)	0.23 [19]	6.30 [20]	0.33 [21]	0.69 [22]	4.84 [23]

and for $\tau \geq 1$

$$f(\tau) = -\frac{\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500}{D} \quad (10)$$

$$\text{Where } D = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right) \quad (11)$$

The parameter, p, which can be thought of as the fraction of the magnetic enthalpy absorbed above the critical temperature, depends on the structure, $p = 0.4$ for BCC_A2 structure and $p = 0.28$ for the others.

For solution phase ϕ (ϕ = Liquid, FCC_A1, RHOMB, HCP_A3, and BCC_A2), T_C^ϕ and β are described by the following expressions:

$$T_c^\phi = x_{Sm}{}^0T_{cSm}^\phi + x_{Ni}{}^0T_{cNi}^\phi + x_{Sm}x_{Ni}L_{Tc}^\phi \quad (12)$$

$$\beta_\phi = x_{Sm}{}^0\beta_{Sm}^\phi + x_{Ni}{}^0\beta_{Ni}^\phi + x_{Sm}x_{Ni}L_\beta^\phi \quad (13)$$

where L_{Tc}^ϕ and L_β^ϕ are the magnetic interaction parameters between elements Sm and Ni. They are set to be zero due to lack of experimental data and extremely low solubility.

For intermetallic compounds, SmNi, $SmNi_2$, $SmNi_3$, $SmNi_5$, and Sm_2Ni_{17}, T_c^ϕ and β^ϕ were taken directly from the measured critical temperature for magnetic ordering (see Table 1). No magnetic data were available for the others intermetallic compounds. As can be seen in Table 1, the Curie temperatures of the intermetallic compounds in the Sm–Ni system, available in the literature, are much below 298.15 K.

3. Literature data

3.1. Phase diagram

The Sm–Ni system has been investigated experimentally by Pan and Cheng [24] and Pan and Nash [25], who determined the phase diagram in the entire composition range, using differential thermal analysis (DTA) with a 10 K/min heating rate and X-ray diffraction (XRD) analysis. According to these authors, SmNi and $SmNi_5$ are congruently melting compounds and Sm_3Ni, $SmNi_2$, $SmNi_3$, Sm_2Ni_7, Sm_5Ni_{19}, and Sm_2Ni_{17} form peritectically. Three eutectic reactions occur in this system (see Table 4). No solubility range was mentioned neither for intermetallic compounds nor for the terminal phases [24,25].

More recently Borzone et al. [26] reinvestigated experimentally the Sm-rich part [50–100 at % Sm] of the Sm–Ni phase diagram using differential scanning calorimetry (DSC) with a slow scanning rate, as well as electron probe microanalysis (EPMA) and X-ray diffraction (XRD). Several experimental difficulties were encountered as a result of the closeness of the thermal effects, undercooling, complex sequences of transformation, and their complex kinetics. Two new phases Sm_7Ni_3 and Sm_3Ni_2 were observed and both of them form through a peritectic reaction (Table 4) and decompose eutectoidly at low temperature. The related invariant reactions and phase relations were determined [26].

Table 3
Thermodynamic parameters of the Sm–Ni binary system.

Phase	Thermodynamic parameters[a]
Liquid	${}^0L_{Sm,Ni} = -164109 + 41.037 \times T$
	${}^1L_{Sm,Ni} = -111489 + 63.932 \times T$
	${}^2L_{Sm,Ni} = 7723 - 9.933 \times T$
RHOMB	${}^0L_{Sm,Ni} = 10^4$
HCP_A3	${}^0L_{Sm,Ni} = 10^4$
BCC_A2	${}^0L_{Sm,Ni} = 10^4$
FCC_A3	${}^0L_{Sm,Ni} = 10^4$
Sm_3Ni	$G_{Sm_3Ni} = 0.75{}^0G_{Sm}^{rhomb} + 0.25{}^0G_{Ni}^{fcc} - 22622 + 3.634 \times T$
Sm_7Ni_3	$G_{Sm_7Ni_3} = 0.70{}^0G_{Sm}^{rhomb} + 0.30{}^0G_{Ni}^{fcc} - 19482 - 3.113 \times T$
Sm_3Ni_2	$G_{Sm_3Ni_2} = 0.60{}^0G_{Sm}^{rhomb} + 0.40{}^0G_{Ni}^{fcc} - 26641 - 1.502 \times T$
SmNi	$G_{SmNi} = 0.50{}^0G_{Sm}^{rhomb} + 0.50{}^0G_{Ni}^{fcc} - 34231 + 0.701 \times T$
$SmNi_2$	$G_{SmNi2} = 0.33{}^0G_{Sm}^{rhomb} + 0.67{}^0G_{Ni}^{fcc} - 35965 + 3.974 \times T$
$SmNi_3$	$G_{SmNi3} = 0.25{}^0G_{Sm}^{rhomb} + 0.75{}^0G_{Ni}^{fcc} - 35441 + 5.806 \times T$
Sm_2Ni_7	$G_{Sm_2Ni_7} = 0.22{}^0G_{Sm}^{rhomb} + 0.78{}^0G_{Ni}^{fcc} - 34780 + 6.328 \times T$
Sm_5Ni_{19}	$G_{Sm_5Ni_{19}} = 0.21{}^0G_{Sm}^{rhomb} + 0.79{}^0G_{Ni}^{fcc} - 34559 + 6.543 \times T$
$SmNi_5$	$G_{SmNi_5} = 0.17{}^0G_{Sm}^{rhomb} + 0.83{}^0G_{Ni}^{fcc} - 31615 + 6.237 \times T$
Sm_2Ni_{17}	$G_{Sm_2Ni_{17}} = 0.10{}^0G_{Sm}^{rhomb} + 0.90{}^0G_{Ni}^{fcc} - 25826 + 8.031 \times T$

[a] Gibbs energies are expressed in (J/mol.at.).

3.2. Thermodynamic data

The enthalpies of formation of Sm_2Ni_7, $SmNi_3$ and $SmNi_2$ estimated by Shilov [27] are −34.322, −35.725, −37.733 kJ/mol of

Table 2
Crystallographic data of the Sm–Ni phases.

Phases	Composition (at. % Ni)	Structure – type	Symbol used in thermo-calc data file	Reference
γ-Sm	0	cI2 – W	BCC_A2	[37]
β-Sm	0	hP2 – Mg	HCP_A3	[38]
α-Sm	0	hR9 – Sm	RHOMB	[39]
Sm_3Ni	25.0	oP16 – Fe_3C	Sm_3Ni	[40]
Sm_7Ni_3	30.0	hP20 – Fe_3Th_7	Sm_7Ni_3	[24]
SmNi	50	oC8 – CrB	SmNi	[41]
$SmNi_2$	66.7	cF24 – Cu_2Mg	$SmNi_2$	[42]
$SmNi_3$	0.75	hR36 – Be_3Nb	$SmNi_3$	[43]
Sm_2Ni_7	77.8	hP36 – $CeNi_7$ HT form	Sm_2Ni_7	[43]
		*hR*54 – Er_2Co_7 LT form		[43]
Sm_5Ni_{19}	79.2	Sm_5Ni_{19} – polytypes	Sm_5Ni_{19}	[36]
$SmNi_5$	83.3	hP6 – $CaCu_5$	$SmNi_5$	[44]
Sm_2Ni_{17}	89.5	hp38 – Th_2Ni_{17}	Sm_2Ni_{17}	[45]
Ni	100	cF4 – Cu	FCC_A1	[46]

Table 4
Invariant reactions in the Sm–Ni phase diagram.

Reaction	Type	Temperature (K)		Composition of the liquid phase (at. % Ni)	
		This work	Literature data [Ref.]	This work	Literature data [Ref.]
Liq ↔ β-Sm + Sm_3Ni	Eutectic	948	948 [26] 937 [24] 936 [35]	22.1	22.0 [26] 26.0 [24] 26.4 [35]
Liq ↔ Sm_3Ni	Congruent	952	951 [26]	–	–
Liq + Sm_3Ni ↔ Sm_7Ni_3	Peritectic	925	927 [26]	32.0	31.0 [26]
Sm_7Ni_3 ↔ Sm_3Ni + Sm_3Ni_2	Eutectoid	890	892 [26]	–	–
Liq ↔ Sm_7Ni_3 + Sm_3Ni_2	Eutectic	896	900 [26]	35.4	35.3 [26]
Liq + SmNi ↔ Sm_3Ni_2	Peritectic	903	902 [26]	35.7	32.0 [26]
Sm_3Ni_2 ↔ Sm_3Ni + SmNi	Eutectoid	873	873 [26]	–	–
Liq ↔ SmNi	Congruent	1345	1352 [24] 1350 [35]	–	–
Liq ↔ SmNi + $SmNi_2$	Eutectic	1094	1082 [24] 1089 [35]	59.3	53.5 [24] 57.8 [35]
Liq + $SmNi_3$ ↔ $SmNi_2$	Peritectic	1297	1304 [24] 1301 [35]	66.2	59.9 [24] 64.6 [35]
Liq + Sm_2Ni_7 ↔ $SmNi_3$	Peritectic	1409	1408 [24,35]	69.1	66.7 [24] 67.8 [35]
Liq + Sm_5Ni_{19} ↔ Sm_2Ni_7	Peritectic	1493	1493 [24]	71.5	68.7 [24]
Liq + $SmNi_5$ ↔ Sm_5Ni_{19}	Peritectic	1555	1555 [24,35]	73.9	71.7 [24] 73.3 [35]
Liq ↔ $SmNi_5$	Congruent	1709	1703 [24] 1706 [35]		
Liq + $SmNi_5$ ↔ Sm_2Ni_{17}	Peritectic	1563	1561 [24] 1565 [35]	90.7	93.5 [24] 91.9 [35]
Liq ↔ Fcc + Sm_2Ni_{17}	Eutectic	1548	1553 [24] 1547 [35]	92.9	94.0 [24] 93.2 [35]

atoms respectively. Using Al solution calorimetry Pasturel et al. [28] measured the enthalpy of formation of $SmNi_5$ and reported a value of −30.3 kJ/mol of atoms. Guo and Kleppa [29,30] measured the enthalpies of formation of SmNi and $SmNi_5$ using direct synthesis calorimetry at high temperature and reported a standard enthalpies of formation of −36.4 ± 0.7 and −27.4 ± 0.5 kJ/mol of atoms respectively.

More recently Borzone et al. [31] investigated the enthalpies of the formation at 300 K of the Sm_3Ni, SmNi and $SmNi_2$ phases by direct calorimetry using two different calorimetric apparatus: aneroid isoperibol calorimeter [32,33] and high temperature direct reaction calorimeter [34]. The values obtained were −21.5 ± 2, −33.0 ± 1 and −36.0 ± 1 kJ/mol of atoms respectively.

The data estimated by Schilov [27] and measured by Pasturel et al. [28] together with the experimental data on the phase diagram determined by Pan and Cheng [24] have been used for a thermodynamic assessment of the Sm–Ni system by means of the Calphad technique carried out by Su et al. [35]. In this optimization, the strongly asymmetric and thermodynamically unlikely liquidus trend around the congruent melting point of SmNi previously suggested, have been discussed. This authors considered that the $SmNi_3$ phase is stable in only at narrow temperature range and fixed arbitrary the value of

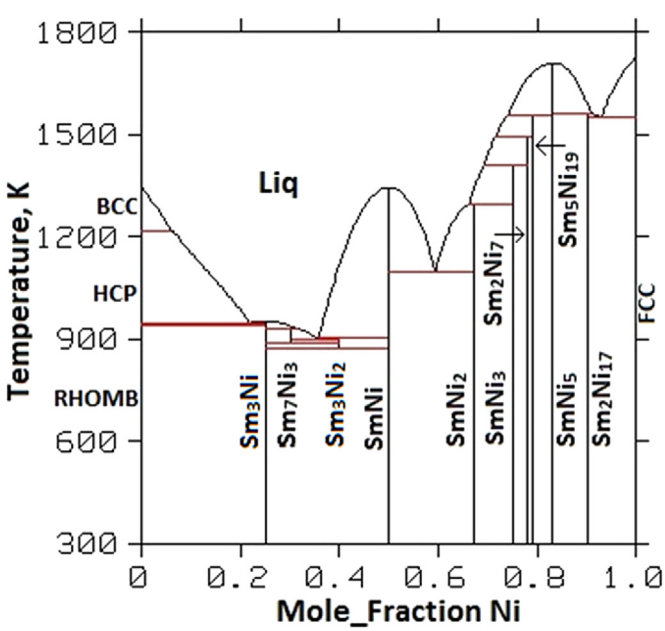

Fig. 1. The Sm–Ni phase diagram calculated from the present thermodynamic description.

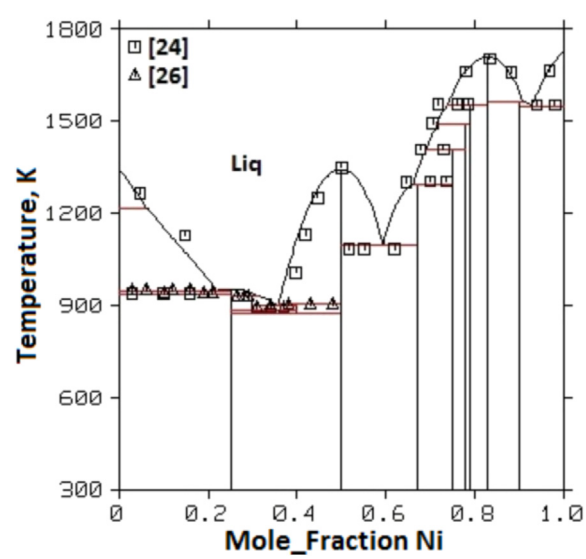

Fig. 2. Calculated phase diagram of Sm–Ni binary system compared with experimental data.

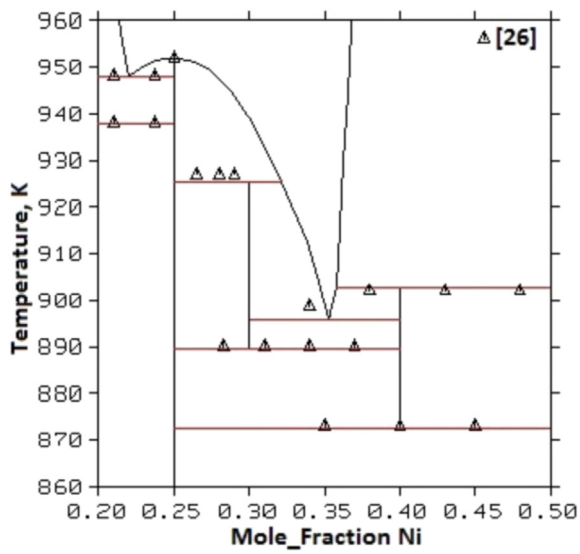

Fig. 3. Enlarged Sm–Ni phase diagram in the Sm–rich part together with experimental data.

entropy of formation to be 1 J/(mol-atoms.K). On the other hand, the calculated enthalpies of formation of the intermetallic compounds are not in good agreement with the experimental data.

3.3. Crystal structures

Crystallographic data of the Sm–Ni phases are listed in Table 2. The Sm_5Ni_{19} structure was studied by Takeda et al. [36] using high resolution transmission electron microscopy and electron diffraction. These authors reported the occurrence of six polytype structures, which have been described as stacking sequences of Laves phase type layers and $CaCu_5$ type layers. No crystal structure relative of the Sm_3Ni_2 phase was reported in the literature.

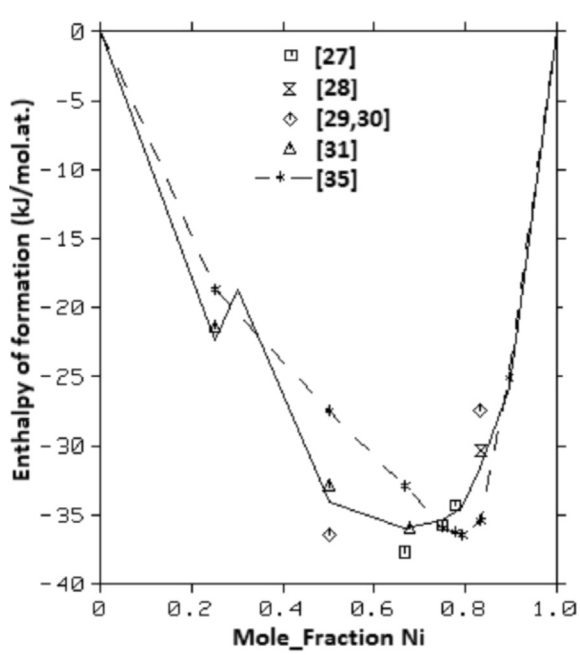

Fig. 4. Calculated enthalpies of formation of Sm–Ni intermetallic compounds at 300 K with literature data.

4. Optimization of thermodynamic parameters

The optimization was carried out by using a computer program Thermocalc. The phase diagram data, and measured enthalpies of compounds were used as input to the program. All the data were first critically reviewed and selected.

On the basis of lattice stabilities cited from Dinsdale [16], the optimization of the model parameters was conducted using the PARROT module [47] in the Thermo-calc software package developed by Sundman et al. [15]. This module works by minimizing the square sum of the differences between experimental data and calculated values. In the optimization procedure, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits. The optimization was carried out by steps. The parameters for the liquid phase were optimized first because the related phase boundaries are readily available. The congruent intermetallic compounds were investigated next. The other parameters for the terminal solid solution phases and intermetallic compounds were consequently optimized by using available phase diagram data and enthalpies of formation of these compounds. All the parameters were finally evaluated together to give the best description of the system. In addition, in the present work solubility was neglected in the FCC_ A1, RHOMB, HCP_A3 and BCC_A2 phases. This was realized by assigning a large positive interaction parameter, i.e.

$$ {}^0L^{RHOMB}_{Sm,Ni} = {}^0L^{HCP_A3}_{Sm,Ni} = {}^0L^{BCC_A2}_{Sm,Ni} = {}^0L^{FCC_A1}_{Sm,Ni} = 10^4 \quad (14) $$

Thermodynamic parameters for all phases in the Sm–Ni binary system obtained in the present work are summarized in Table 3.

Table 4 shows all invariant reactions for this binary system. Reasonable agreement has been realized between the calculated results and experimental data.

According to Dinsdale [16] the stable structure for samarium is rhombohedral at $T < 1190$ K and bcc at $T > 1190$ K, and the hcp structure does not occur at all. In Massalski's book [46], the hcp structure for samarium is stable from 1007 K to 1195 K. Recently, Borzone et al. [26] found experimentally that this structure is stable from 938 K to 1195 K. The energy of transition of Sm from the rhomb structure to the hcp structure is not given by the SGTE database published by Dinsdal [16]. We have hence calculated this energy and found the following expression:

$$ {}^0G^{hcp}_{Sm} = {}^0G^{r\ \mathrm{homb}}_{Sm} + 246.696 - 0.263\,T\ J\ mol^{-1} \quad (15) $$

5. Results and discussion

The calculated phase diagram of the Sm–Ni system is illustrated in Fig. 1. In Fig. 2 we compare our calculated phase diagram with the recent experimental data reported by Borzone et al. [26] and the previous one published by Pan et Cheng [24]. Enlarged Sm-rich part of the calculated Sm–Ni phase diagram is presented in Fig. 3 to show the two new phases recently reported by Borzone et al. [26] and their involved equilibria in a relative narrow range of temperature. The calculated temperature and composition of these equilibria are in good agreement with the experimental ones [26].

In combination with Table 4, reasonable agreement has been

Table 5
Comparison of the measured and optimized enthalpies of formation of the intermetallic compounds of the Sm–Ni system.

Intermetallic compound	Enthalpy of formation (kJ/mol of atoms)	Technique used	T(K)	Reference
Sm_3Ni	−22.6	Optimization	300	This work
	−21.5 ± 2	Direct calorimetry	300	[31]
	−18.7	Optimization		[35]
Sm_7Ni_3	−19.4	Optimization	300	This work
Sm_3Ni_2	−26.6	Optimization	300	This work
SmNi	−34.2	Optimization	300	This work
	−33.0 ± 1	Direct calorimetry	300	[31]
	−36.4 ± 0.7	Direct calorimetry	1473	[30]
	−27.5	Optimization		[35]
$SmNi_2$	−35.9	Optimization	300	This work
	−36.0 ± 1	Direct calorimetry	300	[31]
	−32.9	Optimization		[35]
	−37.7	Prediction		[27]
$SmNi_3$	−35.4	Optimization	300	This work
	−35.9	Optimization		[35]
	−35.7	Prediction		[27]
Sm_2Ni_7	−34.8	Optimization		This work
	−36.3	Optimization		[35]
	−34.3	Prediction		[27]
Sm_5Ni_{19}	−34.5	Optimization	300	This work
	−36.5	Optimization		[35]
$SmNi_5$	−31.6	Optimization	300	This work
	−27.4	Direct calorimetry	1473	[29]
	−30.3	Al colorimetry	300	[28]
	−35.5	Optimization		[35]
Sm_2Ni_{17}	−25.8	Optimization	300	This work
	−25.1	Optimization		[35]

obtained between the calculated and experimental data relative to the invariant reactions in the Sm–Ni system. Most of the calculated reactions are within experimental error. We note that the calculated temperature for the eutectic L ↔ SmNi + $SmNi_2$ at 1082 K deviate about 13 K from the corresponding measured values (1095 K). We consider that this difference is acceptable since the DTA measurement by Pan and Cheng [24] were realized with a heating rate of 10 K/min. Special attention was paid to this equilibrium during the optimization in order to avoid the incompatible asymmetry of the liquidus phase on the left and right sides of the SmNi compound. Indeed, according to Okamoto and Massalski's studies [48,49], judgment about asymmetry of liquidus curves can be made comparing the ratio of the width of the two phase fields on each side of a compound at the same temperature. If the width on one side of the compound is either one half or double that on the other side, the asymmetry may be considered clearly unusual.

Huang et al. [50,51] revised the RE-rich portions (RE = rare earth) of the Pr–Ni and Nd–Ni systems, and proposed a congruent melting for the Pr_7Ni_3 phase, whereas for the Nd_7Ni_3 they proposed a peritectic formation, giving an indication of a progressive different formation behavior of this phase as a function of the R atomic number. Our present results confirm these data and the proposed peritectic formation of the Sm_7Ni_3.

Enthalpies of formation of the Sm–Ni intermetallic compounds were further calculated and compared (Fig. 4) with experimental data published by Borzonne et al. [31] and Guo et Kleppa [29,30]. as well as calculated by Su et al. [35] and those predicted by Shilov [27]. As can be seen in Fig. 4, the calculated enthalpies of formation of $SmNi_3$, $SmNi_2$, SmNi, $SmNi_3$, and $SmNi_5$ agree well with the experimental ones. All values of enthalpies of formation calculated in this work together with the available data from literature are collected in Table 5. We note that the enthalpies of formation calculated by Su et al. [35] are not in good concordance with the experimental values especially for the equiatomic composition for which the deviation is about 5,5 (kJ/mol.at.).

On the other hand, the enthalpy of formation of the $SmNi_2$ intermetallic compound was the most negative. This is in accordance with available RE-Ni_2 (RE = rare earth elements) enthalpies of formation from literature [52,53]. The calculated enthalpies of formation for the Sm_7Ni_3 and Sm_3Ni_2 new phases are less negative than expected since they are stable in a narrow range of high temperatures.

6. Conclusion

The experimental phase equilibria and the thermodynamic data relative to the Gd–Ni system available in the literature have been critically evaluated. Within the scheme of the CALPHAD technique, the thermodynamic models for all the solution phases and intermediate compounds were selected and the Gibbs energy functions were optimized. A set of self-consistent thermodynamic parameters has been obtained. Thermodynamic properties and experimental data especially those relative to the two new phases were well reproduced.

References

[1] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179.
[2] D. Chandra, J.J. Reilly, R. Chellappa, JOM (2006) 26.
[3] K.H.J. Buschow, Mater. Res. Bull. 19 (1984) 935.
[4] S. Sugiomoto, J. Phys. D. Appl. Phys. (2011) 11.
[5] A. Inoue, Prog. Mater. Sci. 43 (1998) 365.
[6] L. Battezzati, S. Pozzovivo, P. Rizzi, Mater Trans. 43 (2002) 2593.
[7] M. Zinkevich, F. Aldinger, Euromat 2003. Symp. T2.
[8] T. Sakai, M. Matsuoka, C. Iwakura, in: K.A. Gschneidner Jr., L. Eyring (Eds.), Handbook on the Physics and Chemistry of Rare Earths, 21, 1995, p. 133.
[9] C. Zimm, A. Jastrab, A. Sternberg, V.K. Pecharsky Jr., K. Gschneidner, M. Osborne, I. Anderson, Adv. Cryog. Eng. 43 (1998) 1759.
[10] C. Zimm, A. Boeder, J. Chell, A. Sternberg, A. Fujita, S. Fujieda, K. Fukamichi, Int. J. Refrig 29 (2006) 1302.
[11] O. Tegus, E. Brück, K.H.J. Buschow, F.R. de Boer, Nature 415 (2002) 150.
[12] M. Balli, O. Sari, C. Mahmed, Ch Besson, Ph Bonhote, D. Duc, J. Forchelet, Appl. Energy 98 (2012) 556.
[13] M. Baricco, M. Palumbo, J. Metall. Nano Mater. 20–21 (2004) 415.
[14] L. Kaufman, H. Bernstein, Computer Calculation of Phase Diagrams, Academic Press, New York, 1970.
[15] B. Sundman, B. Jansson, J.O. Andersson, CALPHAD 9 (1985) 153.

[16] A.T. Dinsdale, CALPHAD 15 (1991) 317.
[17] O. Redlich, A.T. Kister, Ind. Eng. Chem. 40 (1948) 345.
[18] M. Hillert, M. Jarl, CALPHAD 2 (1978) 227.
[19] R.E. Walline, W.E. Wallace, J. Chem. Phys. 6 (41) (1964) 1587.
[20] J. Farrell, W.E. Wallace, J. Inorg. Chem. 5 (1966) 105.
[21] D. Paccard, R. Pauthenet, C.R. Acad. Sci. Paris B 264 (1967) 1056.
[22] R. Ballou, V.M.T.S. Barthem, D. Gignoux, Phys. B C 149 (1988) 340.
[23] J. Laforest, R. Lemaire, D. Paccard, R. Pauthenet, C.R. Acad. Sci. Paris B 264 (1967) 676.
[24] Y.Y. Pan, C.S. Cheng, Acta Phys. Sin. China 32 (1) (1983) 92.
[25] Y.Y. Pan, P. Nash, in: Massalski (Ed.), Phase Diagrams of Binary Nickel Alloys, ASM International, Materials Park, OH, 1991, p. 2861.
[26] G. Borzone, Y. Yuan, S. Delsante, Monatsh Chem. 143 (2012) 1299.
[27] A.L. Shilov, Zh. Fiz. Xhim 61 (5) (1987) 1384.
[28] A. Pasturel, C. Colinet, C. Allibert, P. Hicter, A. Percheron-Guegan, J.C. Achard, Phys. Stat. Sol. b 125 (1984) 101.
[29] Q. Guo, O.J. Kleppa, J. Alloys Compd. 270 (1998) 212.
[30] Q. Guo, O.J. Kleppa, Metall. Mater. Trans. B 29 (1998) 815.
[31] G. Borzone, N. Parodi, R. Raggio, R. Ferro, J. Alloys Compd. 317–318 (2001) 532.
[32] R. Capelli, R. Ferro, A. Borsese, Thermochim. Acta 10 (1974) 13.
[33] R. Ferro, G. Borzone, N. Parodi, G. Cacciamani, J. Phase Equilib. 15 (1994) 317.
[34] G. Cacciamani, G. Borzone, R. Ferro, J. Alloys Compd. 220 (1995) 106.
[35] X. Su, W. Zhang, Z. Du, J. Alloys Compd. 278 (1998) 182.
[36] S. Takeda, Y. Kitano, Y. Komura, J. Less Common Met. 84 (1982) 317.
[37] F.H. Spedding, J.J. Hanak, A.H. Daane, J. Less Common Met. 14 (1961) 110.
[38] J. Kumar, O.N. Srivastava, Acta Crystallogr. B 25 (1969) 2654.
[39] I.R. Harris, J.D. Speight, J. Less Common Met. 114 (1985) 183.
[40] R. Lemaire, D. Paccard, Bull. Soc. Fr. Min. Cristallogr. 90 (1967) 311.
[41] A.E. Dwight, R.A. Conner Jr., J.W. Downey, Acta Crystallogr. 18 (1965) 835.
[42] J.H. Wernick, S. Geller, Trans. AIME 218 (1960) 866.
[43] K.H.J. Buschow, A.S. Van Der Goot, J. Less Common Met. 221 (1970) 419.
[44] K. Nassau, L.V. Cherry, W.E. Wallace, J. Phys. Chem. Solids 16 (1960) 123.
[45] K.H.J. Buschow, J. Less Common Met. 11 (1966) 204.
[46] second ed., in: T.B. Massalski, et al. (Eds.), Binary Alloy Phase Diagrams, vols. 1–3, 1990. Metals Park, OH, USA.
[47] B. Jansson, Tricta-Mac-0234 Royal Institute of Technology, Stockholm, Sweden, 1984.
[48] H. Okamoto, T.B. Massalski, J. Phase Equilib. 14 (3) (1993) 316.
[49] H. Okamoto, T.B. Massalski, J. Phase Equilib. 15 (5) (1994) 500.
[50] M. Huang, D. Wu, K.W. Dennis, J.W. Anderegg, R.W. McCallum, T.A. Lograsso, J. Phase Equilib. Diffus. 26 (2005) 209.
[51] M. Huang, R.W. McCallum, T.A. Lograsso, J. Alloys Compd. 398 (2005) 127.
[52] C. Colinet, A. Pasturel, K.H.J. Buschow, Metall. Trans. 17A (1986) 777.
[53] J. Schott, F. Sommer, J. Less Common Met. 119 (1986) 307.